

Supplementary Material — Part B

Robustness

*What Commentators Talk About: A Validated Thematic Taxonomy of Event-Grounded Sports
Commentary*

Per-cluster evidence behind the four-axis convergence protocol: cross-pipeline survival, the per-cluster axis scores that produce the verdict bands, and the granularity and leave-one-out sensitivity analyses.

Sections in this part

- S3** Algorithm-Axis Per-Cluster Survival Data
- S4** Per-Cluster Four-Axis Validation Scores
- S8** Sensitivity Analysis: MCS Sweep and LOO Pipeline

S3 — Algorithm-Axis Per-Cluster Survival Data

This supplement reports the full per-cluster Algorithm-axis survival data for the seven pre-specified alternative pipelines. Each cell shows the best Hungarian-Jaccard score between the primary cluster and the matched cluster in the alternative pipeline. Green cells (≥ 0.30) count as surviving. Row background colour indicates verdict band. The first section reproduces Table 3 from the main paper for reference.

Pipeline overview (Table 3 from main paper)

Pipeline	DBCV $\uparrow$	Silhouette $\uparrow$	n clusters	Noise %
MiniLM+UMAP+HDBSCAN (primary)	0.348	0.662	34	33.5
MPNet+UMAP+HDBSCAN	0.113	0.308	33	35.1
BGE+UMAP+HDBSCAN	0.237	0.611	35	45.2
GTE+UMAP+HDBSCAN	0.253	0.556	32	41.9
MiniLM+UMAP+KMeans	−0.375	0.440	34	0.0
MiniLM+UMAP+Ward	−0.341	0.420	34	0.0
MiniLM+UMAP+OPTICS	0.234	0.780	36	68.0
MiniLM+UMAP+Leiden	−0.209	0.401	39	0.0

Per-cluster survival matrix

Table 1: Per-cluster Jaccard survival scores across seven alternative pipelines. A cell is green if Jaccard ≥ 0.30 (the Algorithm-axis pass threshold). The rightmost column (N) gives the total survival count.

C#	Label	Size	MPNet	BGE	GTE	OPTICS	KMeans	Ward	Leiden	N
0	DRS-assisted battles	205	0.791	0.741	0.828	1.000	1.000	1.000	1.000	7
1	Time penalties	118	0.442	0.756	0.840	0.483	1.000	1.000	1.000	7
2	Lights-out race starts	54	0.724	0.721	0.767	1.000	1.000	1.000	1.000	7
3	Mirror checks defending	47	0.592	0.667	0.653	1.000	1.000	1.000	1.000	7
4	Race start launch	42	0.653	0.434	0.479	1.000	1.000	0.494	1.000	7
5	Weather conditions	43	0.580	0.623	0.592	1.000	0.524	0.426	1.000	7
6	Grand Prix wins	465	0.269	0.228	0.241	0.329	0.989	0.981	0.989	4
7	Race start lights	36	0.225	0.650	0.234	0.878	0.224	0.222	0.818	3
8	Race control flags	118	0.721	0.019	0.361	0.576	0.733	0.728	0.983	6
9	Formula 1 milestones	134	0.086	0.097	0.128	0.440	0.924	0.957	0.937	4
10	Chicane moves	62	0.038	0.358	0.587	1.000	0.969	1.000	0.984	6
11	Team radio celebrations	55	0.617	0.571	0.613	0.846	0.165	0.166	0.885	5
12	Podium finishes	139	0.527	0.556	0.518	0.403	0.891	0.885	0.946	7
13	Championship & title	158	0.103	0.436	0.250	0.316	0.469	0.476	0.558	5
14	Pit stops	185	0.472	0.515	0.540	0.341	0.907	0.907	0.881	7
15	Tires & compounds	379	0.687	0.675	0.733	0.198	0.959	0.942	0.959	6
16	Safety car deployment	122	0.691	0.645	0.604	0.680	0.931	0.946	0.693	7
17	Gap & pace	67	0.319	0.395	0.432	0.774	0.489	0.660	0.606	7
18	Team dynamics	600	0.410	0.561	0.570	0.000	0.628	0.824	0.792	6
19	Race lead holding	115	0.045	0.306	0.303	0.935	0.426	0.710	0.444	6

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C#	Label	Size	<i>MPNet</i>	<i>BGE</i>	<i>GTE</i>	<i>OPTICS</i>	<i>KMeans</i>	<i>Ward</i>	<i>Leiden</i>	N
20	Wing & car damage	61	0.514	0.245	0.299	0.811	0.540	0.718	0.772	5
21	Braking & lockups	128	0.414	0.329	0.345	0.000	0.465	0.180	0.297	4
22	Wheel-to-wheel	71	0.325	0.440	0.517	0.959	0.696	0.211	0.717	6
23	Circuit action	72	0.033	0.021	0.010	0.761	0.222	0.236	0.136	1
24	Inter-car distance	60	0.025	0.197	0.023	0.833	0.230	0.145	0.137	1
25	Driver & team references	69	0.205	0.051	0.088	0.000	0.212	0.226	0.303	1
26	Defending & attacking	110	0.050	0.266	0.433	0.382	0.341	0.738	0.567	5
27	Overtake battles	762	0.412	0.177	0.493	0.054	0.543	0.299	0.398	4
28	Corner battles	82	0.222	0.230	0.258	0.929	0.364	0.378	0.373	4
29	Corner specific moves	60	0.141	0.174	0.192	0.822	0.267	0.276	0.273	1
30	Lap pace & timing	136	0.069	0.157	0.184	0.000	0.667	0.743	0.586	3
31	Race leader changes	162	0.088	0.181	0.019	0.364	0.658	0.554	0.304	4
32	Front position changes	45	0.105	0.093	0.106	0.000	0.109	0.153	0.174	0
33	Field position changes	122	0.280	0.245	0.261	0.918	0.352	0.364	0.430	4

S4 — Per-Cluster Four-Axis Validation Scores

This supplement tabulates the raw axis scores and binary pass/fail outcomes for all 34 clusters across the four validation axes. Thresholds: A1 (Algorithm) survival fraction $\geq 4/7 \approx 0.571$; A2 (Bootstrap) mean Jaccard ≥ 0.50 ; A3 (Held-out) replication rate ≥ 0.50 ; A4 (Hierarchical) mean Jaccard ≥ 0.50 . Clusters are sorted by verdict band then by size within band.

Table 2: Per-cluster raw scores and pass/fail indicators for all four validation axes. $\checkmark$ = criterion met; $\times$ = criterion not met. A1 score: survival fraction (0–1) across 7 pipelines. A2 score: mean Jaccard across 30 bootstrap subsamples. A3 score: held-out replication rate (rep_strict). A4 score: mean hierarchical Ward Jaccard.

C#	Label	Sz	A1	P	A2	P	A3	P	A4	P	Met	Band
6	Grand Prix wins	465	0.571	$\checkmark$	0.852	$\checkmark$	0.500	$\checkmark$	0.686	$\checkmark$	4	Strict-core
0	DRS-assisted battles	205	1.000	$\checkmark$	0.967	$\checkmark$	0.500	$\checkmark$	0.820	$\checkmark$	4	Strict-core
31	Race leader changes	162	0.571	$\checkmark$	0.635	$\checkmark$	0.500	$\checkmark$	0.606	$\checkmark$	4	Strict-core
9	Formula 1 milestones	134	0.571	$\checkmark$	0.797	$\checkmark$	0.846	$\checkmark$	0.671	$\checkmark$	4	Strict-core
21	Braking & lockups	128	0.571	$\checkmark$	0.579	$\checkmark$	0.600	$\checkmark$	0.532	$\checkmark$	4	Strict-core
16	Safety car deployment	122	1.000	$\checkmark$	0.689	$\checkmark$	0.562	$\checkmark$	0.626	$\checkmark$	4	Strict-core
1	Time penalties	118	1.000	$\checkmark$	0.966	$\checkmark$	0.926	$\checkmark$	0.798	$\checkmark$	4	Strict-core
8	Race control flags	118	0.857	$\checkmark$	0.772	$\checkmark$	0.605	$\checkmark$	0.729	$\checkmark$	4	Strict-core
28	Corner battles	82	0.571	$\checkmark$	0.617	$\checkmark$	0.500	$\checkmark$	0.675	$\checkmark$	4	Strict-core
22	Wheel-to-wheel	71	0.857	$\checkmark$	0.711	$\checkmark$	0.556	$\checkmark$	0.639	$\checkmark$	4	Strict-core
10	Chicane moves	62	0.857	$\checkmark$	0.853	$\checkmark$	1.000	$\checkmark$	0.823	$\checkmark$	4	Strict-core
20	Wing & car damage	61	0.714	$\checkmark$	0.626	$\checkmark$	0.625	$\checkmark$	0.670	$\checkmark$	4	Strict-core
11	Team radio celebrations	55	0.714	$\checkmark$	0.640	$\checkmark$	0.700	$\checkmark$	0.769	$\checkmark$	4	Strict-core
2	Lights-out race starts	54	1.000	$\checkmark$	0.876	$\checkmark$	0.600	$\checkmark$	0.703	$\checkmark$	4	Strict-core
3	Mirror checks defending	47	1.000	$\checkmark$	0.532	$\checkmark$	0.562	$\checkmark$	0.728	$\checkmark$	4	Strict-core
27	Overtake battles	762	0.571	$\checkmark$	0.688	$\checkmark$	0.556	$\checkmark$	0.339	$\times$	3	Robust
18	Team dynamics	600	0.857	$\checkmark$	0.709	$\checkmark$	1.000	$\checkmark$	0.365	$\times$	3	Robust
15	Tires & compounds	379	0.857	$\checkmark$	0.827	$\checkmark$	0.385	$\times$	0.575	$\checkmark$	3	Robust
14	Pit stops	185	1.000	$\checkmark$	0.776	$\checkmark$	0.667	$\checkmark$	0.447	$\times$	3	Robust
13	Championship & title	158	0.714	$\checkmark$	0.693	$\checkmark$	0.600	$\checkmark$	0.492	$\times$	3	Robust
12	Podium finishes	139	1.000	$\checkmark$	0.826	$\checkmark$	0.462	$\times$	0.767	$\checkmark$	3	Robust
30	Lap pace & timing	136	0.429	$\times$	0.697	$\checkmark$	0.900	$\checkmark$	0.534	$\checkmark$	3	Robust
33	Field position changes	122	0.571	$\checkmark$	0.510	$\checkmark$	0.619	$\checkmark$	0.392	$\times$	3	Robust
19	Race lead holding	115	0.857	$\checkmark$	0.694	$\checkmark$	0.300	$\times$	0.620	$\checkmark$	3	Robust
5	Weather conditions	43	1.000	$\checkmark$	0.075	$\times$	0.667	$\checkmark$	0.716	$\checkmark$	3	Robust
4	Race start launch	42	1.000	$\checkmark$	0.358	$\times$	0.600	$\checkmark$	0.861	$\checkmark$	3	Robust
26	Defending & attacking	110	0.714	$\checkmark$	0.664	$\checkmark$	0.333	$\times$	0.481	$\times$	2	Moderate
23	Circuit action	72	0.143	$\times$	0.700	$\checkmark$	0.400	$\times$	0.585	$\checkmark$	2	Moderate
17	Gap & pace	67	1.000	$\checkmark$	0.461	$\times$	0.333	$\times$	0.602	$\checkmark$	2	Moderate
7	Race start lights	36	0.429	$\times$	0.041	$\times$	0.636	$\checkmark$	0.864	$\checkmark$	2	Moderate
29	Corner specific moves	60	0.143	$\times$	0.445	$\times$	0.375	$\times$	0.717	$\checkmark$	1	Weak
25	Driver & team references	69	0.143	$\times$	0.472	$\times$	0.455	$\times$	0.481	$\times$	0	Fragile
24	Inter-car distance	60	0.143	$\times$	0.411	$\times$	0.417	$\times$	0.475	$\times$	0	Fragile
32	Front position changes	45	0.000	$\times$	0.241	$\times$		$\times$	0.407	$\times$	0	Fragile

S8 — Sensitivity Analysis: MCS Sweep and LOO Pipeline

This supplement reports two pre-specified sensitivity analyses. Section 1 shows the effect of varying `min_cluster_size` (mcs) from 15 to 120 on key clustering metrics (cluster count, noise rate, DBCV, silhouette, vocabulary richness); the mcs = 10 point cited in Section 4.2 is from a separate extension run. The chosen value mcs = 35 (highlighted) sits at the centre of a stability plateau. Section 2 reports leave-one-out sensitivity for the Algorithm axis: how much the 27/34 majority-threshold count shifts when each of the seven alternative pipelines is removed.

1. min_cluster_size sensitivity sweep

Table 3: Sensitivity of the clustering pipeline to `min_cluster_size` (mcs). The highlighted row (mcs = 35) is the value used in the main analysis. DBCV: Density-Based Cluster Validation (Moulávi et al., 2014). Distinct. terms: mean distinctive unigrams per cluster under the sweep’s screen (log-odds > 2.0 at frequency ≥ 3) – a cheap ranking statistic, *not* the FDR-corrected R3 count of Section 4.3 (25.4 at mcs = 35); the scales are not comparable. Interp: clusters with ≥ 5 such terms.

mcs	<i>n</i> clusters	Noise	DBCV	Silhouette	Distinct. terms	Interp.
15	62	35.2%	0.353	0.633	11.3	89%
20	44	30.4%	0.346	0.640	13.8	91%
25	38	29.4%	0.303	0.635	15.3	97%
30	35	34.4%	0.360	0.667	15.3	94%
35	34	33.5%	0.348	0.662	17.2	100%
40	33	34.2%	0.322	0.669	16.7	97%
45	30	34.6%	0.316	0.658	16.8	100%
50	29	35.7%	0.328	0.660	17.0	100%
55	3	0.0%	0.360	0.347	18.7	100%
60	3	0.0%	0.360	0.347	18.7	100%
70	3	0.0%	0.360	0.347	18.7	100%
80	3	0.0%	0.360	0.347	18.7	100%
100	3	0.0%	0.360	0.347	18.7	100%
120	2	0.0%	0.332	0.494	15.5	50%

2. Leave-one-out pipeline sensitivity

The full Algorithm-axis majority count (all 7 judges) is 27/34. The table below shows how this count changes when each pipeline is removed from the judge panel (6 remaining pipelines; majority threshold $\geq 4/6$). No single pipeline is disproportionately influential.

Table 4: Leave-one-out (LOO) sensitivity: Algorithm-axis majority-threshold count when each pipeline is removed in turn. Counts are noise-adjusted, matching the 27/34 Algorithm-axis figure reported in the main text; full count (all 7 judges) = 27/34, Δ = change from 27.

Removed pipeline	Majority count	Δ
OPTICS	22	-5
Ward	22	-5
KMeans	23	-4
A1_MPNNet	26	-1
A2_BGE	26	-1
A3_GTE	25	-2
B2_Leiden	25	-2